

CSE 4109

Topic: Multi-Agent Search and Adversarial Environments

Selected paper: Silver, D., Schrittwieser, J., Simonyan, K. *et al.* Mastering the game of Go without human knowledge. *Nature* **550**, 354–359 (2017).

<https://doi.org/10.1038/nature24270>

What Problem the Authors Are Trying to Solve

The authors are seeking to tackle a fundamental issue in artificial intelligence: Can a computer achieve mastery in a complex strategic game without any human input or knowledge?

How the authors approached or solved the problem

Three core elements work together synergistically and produce a very strong synergy for AlphaGo Zero:

Neural Net: The neural net receives an image of the board as input. The output contains two items: (1) a Value (-1 to +1) indicating how well the position is performing on the board (who is winning); and (2) a Policy that represents the probability distribution of the moves the AI may choose to make from that position. The Integrated Neural Net replaces the use of handcrafted value functions.

Monte Carlo Tree Search (or MCTS) : evaluates each move with an AI performing 1,600 simulations before progressing through the four stages (Selection, Expansion, Evaluation, Backup) of the tree. The MCTS method balances exploration and exploitation through the use of an Upper Confidence Bounds (UCB) method.

Self-Play Training Cycle:

- The current version of the AI plays against itself using the MCTS.
- The current version maintains records of all positions (in its Calendar Database) and refines the MCTS with the result of each match.
- The current version trains the AI to predict the future winner and align to the predicted Optimal Probability as determined by MCTS.
- The new version of the AI MUST win over 55% of the matches versus an older version to replace it.

In conclusion:

- AlphaGo Zero required just 4 TPUs to run—compared to the original AlphaGo Lee's 48 TPUs—a resource saving of 12x.
- AlphaGo Zero trained faster than AlphaGo Lee to reach superior play (3 days of training).
- Final Elo rating for AlphaGo Zero was 5,185 versus 3,739 for AlphaGo Lee.
- AlphaGo Zero defeated AlphaGo Lee in ALL matches played (100-0) on March 24, 2023, with the original AlphaGo having previously defeated the World Champion.

How Do Course Topics Relate to AlphaGo Zero?

The Traditional Techniques include the following:

- Minimax, which analyzes the game's tree assuming both players will choose the best available action.
- Alpha-Beta Pruning, which reduces the number of branches from the game tree that do not change the final outcome.
- Evaluation Functions, which are heuristic evaluations made manually on possible moves in a position.
- Informed Search, which searches from a starting point toward an end point using a heuristic to direct the search.

- Uninformed Search, which searches systematically without having any prior knowledge of the game.

Similarities Between AlphaGo Zero and These Traditional Techniques:

- Both AlphaGo Zero and the traditional techniques use principles of tree searching.
- Both types of systems must compete against an opponent trying to win as well.
- In both systems, some type of evaluation must be made about the position.
- Both systems must set a limit to how many layers deep they will search, based on system constraints.

Differences Between AlphaGo Zero and Traditional Techniques:

- **Search Strategy:** AlphaGo Zero uses a form of selective sampling based on statistics, while the traditional techniques use a systematic approach across the tree.
- **Evaluation:** AlphaGo Zero uses the technique of using neural networks for training, while traditional techniques create functions for the evaluation based on manual input from experts.
- **Source of Knowledge:** AlphaGo Zero generates its own knowledge by self-playing, while the traditional techniques generate the knowledge through programming by experts in the game.
- **Branching Factor:** The AlphaGo Zero policy network focuses on evaluating five to ten candidate moves, while traditional techniques evaluate all the possible moves with pruning.

Improvements

The number of moves available in a given position in Go is approximately 250, which is too many for traditional techniques to handle effectively; however, the usage of a policy network significantly reduces the effective branching factor; additionally, the evaluation learned through AlphaGo Zero captures subtle aspects of many positions that humans find difficult to describe.

MCTS makes real time assessments of the current complexity of the game and assigns an amount of time, or budget, to the analysis of a position.

AlphaZero was able to be trained at 10 times the speed of what we would expect from a “traditional” AI, its neural network has increased operational efficiency by a factor of 12,

with AlphaZero outperforming any other AI at that time by more than 1400 points on the Elo Rating Scale.

It was shown, independently from Mnuchin AI, that AI has the ability to find and make connections independently to solve problems in minutes-hundreds of years of experience and knowledge for humans, and as such could potentially apply to areas where humans have less knowledge/skill.

AI does not make use of substandard tactics or have the inherent human-biased tendencies.

Reducing complexity improves the ability of AI to produce results via AlphaZero.

The same algorithm was used (AlphaZero) to teach shogi (Japanese Chess) and Chess (Western Chess), without any bespoke engineering to develop the algorithms specifically for each game.

My Suggestion

- Competition in a Tournament-style: In order to help improve the training process of your model, it helps to give your model as many different games as possible so that your model has access to a lot of data to identify trends across those games and learn how to quickly learn about new games.
- Our stripped down product line will be priced significantly lower and easier to use than our traditional line of products for those users who do not require as much computing power. As such, we will develop lightweight versions of our flagship line of products, which will fall between 10 to 100 times smaller than our current flagship models within the 2,000 to 3,000 dollar price range.
- Like an expert, observe what strategies your opponent employs when playing against them as a human player, and adjust your strategy to remain competitive.
- Trust is Built Through Transparency: The relationship that builds between a user and AI based on transparency is created by allowing the user to see how the AI derived its strategy visually.

- Providing a model with a broader range of agents during training will enable your model to learn a variety of strategies for each game and create a significantly greater opportunity for the exploration of different forms of play.

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